Trainings

Install

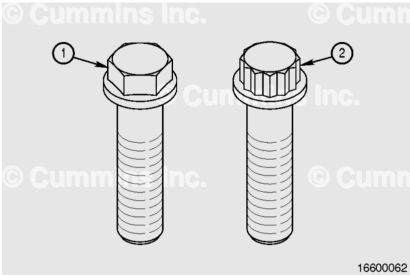
Note : There are two types of flywheel mounting capscrews used on the QSK45 and QSK60 engines.

The capscrews can be identified by having either a 6-point head or a 12-point head. Engines using the 12-point head capscrews also have a clamping ring installed.

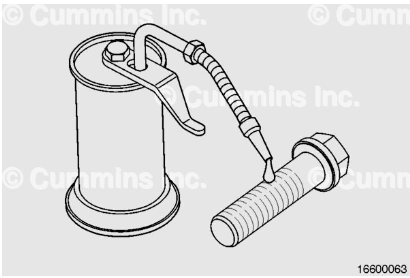
1. 6-point head capscrew
2. 12-point head capscrew.

Note : The 12-point head capscrew is only used on the QSK45 engine in certain excavator applications.

Note : For the QSK45 excavator applications that use the 12-point head capscrews (2), remove and discard the flywheel, capscrews, and spacer at end of every life rebuild.

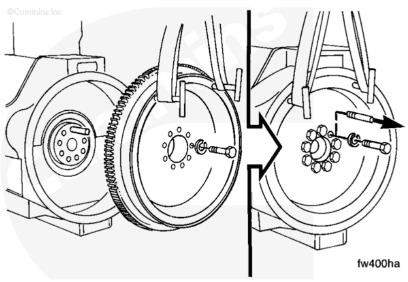


Lubricate the flywheel mounting capscrew threads and flange head surfaces with clean engine oil.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ WARNING ⚠

Install a new roll pin into the dowel hole in the rear face of the crankshaft to achieve the correct protrusion, 8.7mm +/- 1.0mm.

Install two guide studs to prevent the flywheel from rotating.

Install the flywheel and line up the dowel hole in the flywheel with the roll pin in the crankshaft.

Install the flywheel mounting capscrews.

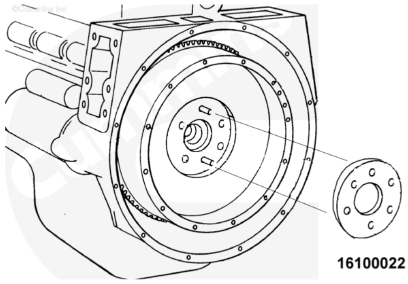
Remove the two guide studs and install the last two flywheel mounting capscrews.

Remove the two T-handles.

Note : Perform this step only for QSK45 applications that use the 12-point head capscrews and clamping ring.

Insert the clamping ring against the flywheel. The clamping ring can be installed with either side toward the flywheel.

Install the 12-point head flywheel mounting capscrews.

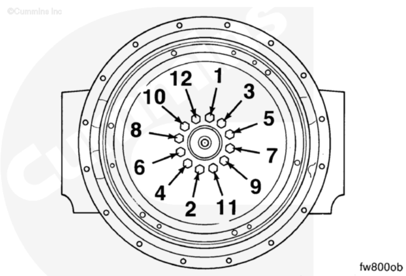


For the 6-point head capscrews:

Tighten the flywheel mounting capscrews in the sequence shown.

Torque Value:

1. 200 n•m [148 ft-lb]
2. 380 n•m [280 ft-lb]
3. 685 n•m [505 ft-lb]



For the 12-point head capscrews:

Tighten the flywheel mounting capscrews in the sequence shown.

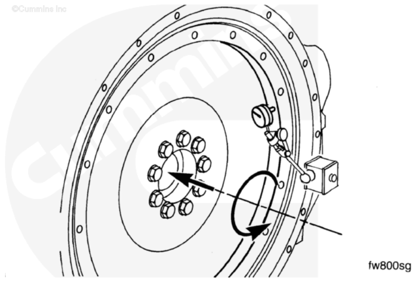
Torque Value:

1. 200 n•m [148 ft-lb]
2. 380 n•m [280 ft-lb]
3. 685 n•m [505 ft-lb]
4. 800 n•m [590 ft-lb]

Measure the flywheel total indicator runout.

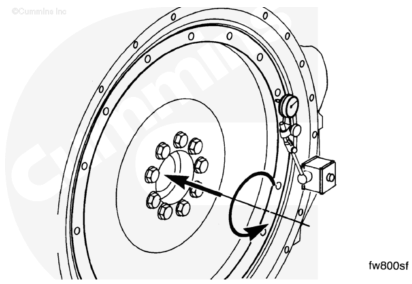
Measurements			
		mm	in
Flywheel Housing	Number 0	648	25.5
	Number 00	787	31.0

Flywheel Bore Total Indicator Runout				
	mm		in	
Number 0		0.127	MAX	0.005
Number 00		0.127	MAX	0.005



The maximum allowable flywheel face runout is determined by the distance from the center of the flywheel to the contact point of measurement, divided by 1000.

Measure the flywheel face total runout.



Flywheel Face Total Indicator Runout				
	mm		in	
Number 0 (with a 254 mm [10 in] center distance)		0.254	MAX	0.010
Number 00 (with a 305 mm [12 in] center distance)		0.305	MAX	0.012